

SQL Queries

Exercise 5.1

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- Consider the following relations:

Student (*snum*: integer, *sname*: string, *major*: string, *level*: string, *age*: integer)

Class (*name*: string, *meets at*: string, *room*: string, *fid*: integer)

Enrolled (*snum*: integer, *cname*: string)

Faculty (*fid*: integer, *fname*: string, *deptid*: integer)

The meaning of these relations is straightforward; for example, Enrolled has one record per student-class pair such that the student is enrolled in the class. Write the following queries in SQL. No duplicates should be printed in any of the answers.

Find the names of all Juniors (level = JR) who are enrolled in a class taught by I. Teach.

```
SELECT DISTINCT S.Sname
FROM Student S, Class C, Enrolled E, Faculty F
WHERE S.snum = E.snum AND
      E.cname = C.name AND
      C.fid = F.fid AND
      F.fname = 'I.Teach' AND
      S.level = 'JR'
```

Student (snum, sname, major, level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the age of the oldest student who is either a History major or enrolled in a course taught by I. Teach.

```
SELECT MAX(S.age)
FROM Student S
WHERE (S.major = 'History')
      OR S.snum IN (SELECT E.snum
                    FROM Class C, Enrolled E, Faculty F
                    WHERE E.cname = C.name AND C.fid = F.fid
                      AND F.fname = 'I.Teach' )
```

Student (snum, sname, major, level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the names of all classes that either meet in room R128 or have five or more students enrolled.

```
SELECT C.name
FROM Class C
WHERE C.room = 'R128'
       OR C.name IN (SELECT E.cname
                     FROM Enrolled E
                     GROUP BY E.cname
                     HAVING COUNT (*) >= 5)
```

Student (snum, sname, major, level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the names of all students who are enrolled in two classes that meet at the same time.

```
SELECT DISTINCT S.sname
FROM Student S
WHERE S.snum IN (SELECT E1.snum
                 FROM Enrolled E1, Enrolled E2, Class C1, Class C2
                 WHERE E1.snum = E2.snum AND
                      E1.cname <> E2.cname AND
                      E1.cname = C1.name AND
                      E2.cname = C2.name AND
                      C1.meets at = C2.meets at)
```

Student (snum, sname, major, level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the names of faculty members who teach in every room in which some class is taught.

```
SELECT DISTINCT F.fname
FROM Faculty F
WHERE NOT EXISTS (( SELECT C.room
                     FROM Class C )
                  EXCEPT
                  (SELECT C1.room
                   FROM Class C1
                   WHERE C1.fid = F.fid ))
```

Student (snum, sname, major,
level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the names of faculty members for whom the combined enrollment of the courses that they teach is less than five.

```
SELECT DISTINCT F.fname
FROM Faculty F
WHERE 5 > (SELECT COUNT(E.snum)
           FROM Class C, Enrolled E
           WHERE C.name = E.cname
                AND C.fid = F.fid)
```

Student (snum, sname, major,
level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

For each level, print the level and the average age of students for that level.

```
SELECT S.level, AVG(S.age)
FROM Student S
GROUP BY S.level
```

Student (snum, sname, major, level, age)
Class (name, meets at, room, fid)
Enrolled (snum, cname)
Faculty (fid, fname, deptid)

For all levels except JR, print the level and the average age of students for that level.

```
SELECT S.level, AVG(S.age)
FROM Student S
WHERE S.level <> 'JR'
GROUP BY S.level
```

Student (snum, sname, major,
level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

For each faculty member that has taught classes only in room R128, print the faculty member's name and the total number of classes she or he has taught.

```
SELECT F.fname, COUNT(*) AS CourseCount
FROM Faculty F, Class C
WHERE F.fid = C.fid
GROUP BY F.fid, F.fname
HAVING EVERY ( C.room = 'R128' )
```

Student (snum, sname, major, level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the names of students enrolled in the maximum number of classes.

```
SELECT DISTINCT S.sname
FROM Student S
WHERE S.snum IN (SELECT E.snum
                  FROM Enrolled E
                  GROUP BY E.snum
                  HAVING COUNT (*) >= ALL (SELECT COUNT (*)
                                           FROM Enrolled E2
                                           GROUP BY E2.snum ))
```

Student (snum, sname, major, level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

Find the names of students not enrolled in any class.

```
SELECT DISTINCT S.sname  
FROM Student S  
WHERE S.snum NOT IN (SELECT E.snum  
                     FROM Enrolled E )
```

Student (snum, sname, major,
level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)

For each age value that appears in Students, find the level value that appears most often. For example, if there are more FR level students aged 18 than SR, JR, or SO students aged 18, you should print the pair (18, FR).

```
SELECT S.age, S.level
FROM Student S
GROUP BY S.age, S.level,
HAVING S.level IN (SELECT S1.level
                   FROM Student S1
                   WHERE S1.age = S.age
                   GROUP BY S1.level, S1.age
                   HAVING COUNT (*) >= ALL (SELECT COUNT (*)
                                           FROM Student S2
                                           WHERE S1.age = S2.age
                                           GROUP BY S2.level, S2.age))
```

Student (snum, sname, major,
level, age)

Class (name, meets at, room, fid)

Enrolled (snum, cname)

Faculty (fid, fname, deptid)